Notebook 3 — Chapter 3.3

Selection Bias, Adverse Selection & Information Economics

20 worked examples — conditioning on execution, winner's curse, trading against asymmetric information — calibrated for competition trading intuition

A. Conditioning on Trade Execution

Example 1 — The Core Idea: a Trade Is Not a Random Sample

True value $V = \$100$ w.p. 0.5, $V = \$90$ w.p. 0.5 (so $\mathbb{E}[V] = 95$). An informed trader (who knows V exactly) buys from you only when V exceeds your ask of \$96. Find $\mathbb{E}[V \mid \text{trade executed}]$.
Solution. The informed trader buys iff $V = 100 > 96$; they never buy when $V = 90$. So a trade occurring tells you $V = 100$ with certainty:

$$\mathbb{E}[V \mid \text{trade}] = \boxed{\$100} \neq \mathbb{E}[V] = \$95$$

⋆ *Intuition:* this is the entire chapter in one line. A fill against an informed counterparty is evidence *about* V, not a neutral event — exactly the mechanism previewed in Chapter 3.2's Example 11.

Example 2 — Conditioning Selects a Subset

V is equally likely to be $\$80, 90, 100, 110, 120$ (so $\mathbb{E}[V] = 100$). You quote ask = \$105; an informed trader buys iff $V > 105$. Find $\mathbb{E}[V \mid \text{buy}]$.
Solution. Only $V \in \{110, 120\}$ triggers a buy:

$$\mathbb{E}[V \mid \text{buy}] = \frac{110 + 120}{2} = \boxed{\$115}$$

▶ *Speed trick:* conditioning on a threshold just means "average over the surviving outcomes only" — list which outcomes survive first, then average, never average all outcomes and adjust after.

Example 3 — Noise Trading Dilutes the Curse

Same $V \in \{80, 90, 100, 110, 120\}$ (equal prob.), ask = \$105. Half of all buy orders are *noise* (buy regardless of V); half are *informed* (buy iff $V > 105$). Find $\mathbb{E}[V \mid \text{buy order arrives}]$.
Solution. Weight each V by its probability of generating *either* type of buy: weight $= 0.5(1) + 0.5 \cdot \mathbb{I}(V > 105)$. For $V \leq 100$: weight = 0.5. For $V \geq 110$: weight = 1.0.

$$\mathbb{E}[V \mid \text{buy}] = \frac{0.5(80{+}90{+}100) + 1.0(110{+}120)}{0.5(3) + 1.0(2)} = \frac{135 + 230}{3.5} \approx \boxed{\$104.29}$$

⋆ *Intuition:* compare to Example 2's \$115 (100% informed) — noise trading *dilutes* the adverse-selection signal. This is exactly why spreads can be tighter in more liquid, noise-heavy markets: each individual fill is less informative.

Example 4 — The Complementary Event: No Trade

In Example 1's setup ($V \in \{90, 100\}$, informed buys iff $V > 96$), find $\mathbb{E}[V \mid \text{no trade}]$.
Solution. No trade happens exactly when $V = 90$ (the informed trader stays away):

$$\mathbb{E}[V \mid \text{no trade}] = \boxed{\$90}$$

▶ *Intuition:* "no fill" carries information too — silence from an informed counterparty, when you'd expect them to trade if V were high, is itself (weak) evidence V is low. Markets read the absence of flow, not just its presence.

Example 5 — The Continuous, General-Purpose Formula

$V \sim \text{Unif}(0, 100)$. You quote ask $= a$; an informed trader buys iff $V > a$. Find $\mathbb{E}[V \mid \text{buy}]$ as a function of a, and evaluate at $a = 60$.
Solution. Conditional on $V > a$, $V \sim \text{Unif}(a, 100)$ (a direct order-statistics-style truncation, Ch. 2.3), so:

$$\mathbb{E}[V \mid \text{buy}] = \frac{a + 100}{2}$$

At $a = 60$:

$$\mathbb{E}[V \mid \text{buy}] = \frac{60 + 100}{2} = \boxed{\$80}$$

⋆ *Speed trick:* for a uniform V, conditioning on a threshold always gives the midpoint of the surviving interval — memorize this shape, it reappears throughout Sections B and D.

B. Winner's Curse

Example 6 — The Classical Jar-of-Coins Setup

True (unknown, fixed) value V. $n = 5$ bidders each see $V + \text{noise}_i$, $\text{noise}_i \overset{\text{iid}}{\sim} \text{Unif}(-10, 10)$, and bid their own estimate naively. The winner's estimate is $V + \max_i(\text{noise}_i)$. Find the average overstatement of V by the winning bid.
Solution. Using the order-statistic mean for $\text{Unif}(a, b)$ with n draws, $\mathbb{E}[\max] = a + (b - a)\frac{n}{n+1}$:

$$\mathbb{E}[\max_i \text{noise}_i] = -10 + 20 \cdot \frac{5}{6} = \boxed{\$6.67}$$

⋆ *Intuition:* the winner isn't the bidder who happened to value the item most accurately — they're the bidder whose *noise* happened to be most positive. Winning is itself evidence your estimate is too high.

Example 7 — The Shading Correction

Using Example 6's result, if your raw estimate is \$250, what should you actually bid to correct for the winner's curse (as a practical first-order approximation)?
Solution. Shade down by the expected overstatement computed in Example 6:

$$\text{bid} = 250 - 6.67 = \boxed{\$243.33}$$

▶ *Intuition:* this is an approximation (the *exact* Bayesian-Nash shading conditions on "my noise is the max," not just "the max of n draws"), but it captures the right order of magnitude and the right direction — always shade down, never bid your raw estimate in a common-value auction.

Example 8 — More Bidders, Worse Curse

Noise $\sim \text{Unif}(-5, 5)$. Compare the expected winner's curse at $n = 2$ bidders vs. $n = 10$ bidders.
Solution.

$$n = 2: \ -5 + 10 \cdot \frac{2}{3} \approx \boxed{\$1.67} \qquad n = 10: \ -5 + 10 \cdot \frac{10}{11} \approx \boxed{\$4.09}$$

⋆ *Intuition:* more competition doesn't just mean a higher expected winning price — it means the winning estimate is drawn from a more extreme order statistic, so the curse itself grows. A hot, crowded auction needs *more* bid-shading, not less — the opposite of naive intuition.

Example 9 — Winner's Curse for a Passive Limit Order

$V \sim \text{Unif}(0, 100)$. You rest a passive limit *buy* order at $p = \$40$. An informed counterparty sells to you only when $V \leq p$ (i.e., only when your bid is generous relative to true value). Find $\mathbb{E}[V \mid \text{filled}]$.
Solution. Filled iff $V \leq 40$, so $V \mid \text{fill} \sim \text{Unif}(0, 40)$:

$$\mathbb{E}[V \mid \text{fill}] = \boxed{\$20}$$

▶ *Intuition:* this is Example 1's mechanism, mirror-imaged — *getting filled* on a passive order is the winner's curse for market makers. Just like winning an auction is bad news about the item, getting your resting bid hit is bad news about V.

Example 10 — Self-Consistent Quoting

Same setup as Example 9. Find the limit buy price p such that your post-fill expected edge is exactly \$5, i.e. $\mathbb{E}[V \mid \text{fill}] = p - 5$.
Solution. $\mathbb{E}[V \mid \text{fill}] = p/2$ (Example 5's uniform-midpoint shape). Solve:

$$\frac{p}{2} = p - 5 \ \Rightarrow \ \frac{p}{2} = 5 \ \Rightarrow \ p = \boxed{\$10}$$

⋆ *Intuition:* to guarantee a \$5 edge *after* conditioning on the fill, you must quote far below the naive fair value (\$50) — quoting at \$10 feels absurdly conservative until you remember every fill here is, by construction, adverse.

C. Trading Against Asymmetric Information

Example 11 — Bayesian Updating on a Trade Signal

Prior: $V = 100$ w.p. 0.6, $V = 80$ w.p. 0.4 ($\mathbb{E}[V] = 92$). Informed traders buy w.p. 0.9 if $V = 100$, w.p. 0.1 if $V = 80$. You observe a buy order. Update your belief.
Solution. By Bayes' rule:

$$\mathbb{P}(V = 100 \mid \text{buy}) = \frac{0.6(0.9)}{0.6(0.9) + 0.4(0.1)} = \frac{0.54}{0.58} \approx 0.931$$

$$\mathbb{E}[V \mid \text{buy}] = 100(0.931) + 80(0.069) \approx \boxed{\$98.62}$$

▶ *Intuition:* this is Section A's conditioning logic made fully quantitative

via Bayes' rule — the belief moved from 92 to 98.62, a direct, computable consequence of "buys are more likely when V is high."

Example 12 — Re-Quoting After the Update

Before the buy order in Example 11, you quoted ±\$1 around the prior mean (91/93). After observing the buy, requote ±\$1 around your updated belief.

Solution. New center $= \mathbb{E}[V \mid \text{buy}] = 98.62$:

$$\text{bid} = \boxed{\$97.62}, \quad \text{ask} = \boxed{\$99.62}$$

⋆ *Intuition:* a fill doesn't just change your inventory (Ch. 3.2) — it changes your *beliefs* for the next quote. Both effects push the same direction here (a buy fill means both "you're now short" and "V is probably higher"), compounding the skew.

Example 13 — Order Size as a Signal

Prior: $V = 100$ or 80, equal probability. Informed traders buy only when $V = 100$; noise traders buy regardless of V. Large orders are informed w.p. 0.7; small orders are informed w.p. 0.2. Find $\mathbb{E}[V \mid \text{buy}]$ for each order size.

Solution. Both types buy w.p. 1 when $V = 100$ (so $\mathbb{P}(\text{buy}, V{=}100) = 0.5$ regardless of size); only noise buys when $V = 80$.

$$\text{Large: } \mathbb{P}(\text{buy}, 80) = 0.5(0.3) = 0.15 \Rightarrow \mathbb{E}[V \mid \text{buy}] = \frac{50+12}{0.65} \approx \boxed{\$95.38}$$

$$\text{Small: } \mathbb{P}(\text{buy}, 80) = 0.5(0.8) = 0.40 \Rightarrow \mathbb{E}[V \mid \text{buy}] = \frac{50+32}{0.90} \approx \boxed{\$91.11}$$

▶ *Intuition:* both numbers exceed the prior mean of \$90, but the large order is far more contaminated with informed flow — "size is information" is one of the most robust, real empirical facts in market microstructure.

Example 14 — Size-Dependent Defensive Quoting

Using Example 13, set asks for large and small orders targeting a flat \$0.50 edge above each conditional mean.

Solution.

$$\text{ask}_{\text{large}} = 95.38 + 0.50 = \boxed{\$95.88}$$

$$\text{ask}_{\text{small}} = 91.11 + 0.50 = \boxed{\$91.61}$$

⋆ *Intuition:* a market maker who quotes the *same* price regardless of order size is leaving themselves exposed on large orders and overcharging on small ones — real venues price size tiers precisely because of Example 13's logic.

Example 15 — Naming the Model: Kyle's Linear Equilibrium

An informed trader with signal $\delta \sim \mathcal{N}(0, \sigma_v^2{=}1)$ submits order size $x = \beta\delta$; noise traders submit $u \sim \mathcal{N}(0, \sigma_u^2{=}4)$; the market maker sets $p = \lambda(x+u)$. Kyle's (1985) equilibrium gives $\beta = \sigma_u/\sigma_v$, $\lambda = \sigma_v/(2\sigma_u)$. Compute β, λ, and the price impact of $\delta = 0.5$ assuming zero noise that round.

Solution.

$$\beta = \frac{2}{1} = 2, \qquad \lambda = \frac{1}{2(2)} = 0.25$$

$$x = \beta\delta = 2(0.5) = 1, \qquad p = \lambda(x+0) = 0.25(1) = \boxed{0.25} = \delta$$

▶ *Intuition:* with noise = 0, the price exactly reveals the signal ($p = \delta$) — in general noise trading is what allows the informed trader to hide *at all*. This is the named, canonical model behind everything in Section C; worth recognizing by name even without re-deriving β, λ from scratch.

D. Capstones

Example 16 — The Breakeven Spread Against Mixed Flow

A fraction $p = 0.3$ of counterparties are informed, extracting \$2 from you per trade; the rest are noise, contributing some edge e to you per trade. Find the minimum e for you to break even overall.

Solution. Breakeven: $(1-p)e = p(2)$:

$$e = \frac{0.3(2)}{0.7} \approx \boxed{\$0.857}$$

⋆ *Intuition:* if your current quoted edge from noise flow is below this threshold (e.g. \$0.50), you are losing money on net even though every *individual* noise trade looks profitable — the informed losses are concentrated and must be covered by the rest of the book, not any single trade.

Example 17 — Sequential Updating: Two Buys in a Row

Continuing Example 11, a *second* buy order arrives (same likelihoods, applied to the already-updated belief $\mathbb{P}(V{=}100) = 0.931$). Update again.

Solution.

$$\mathbb{P}(V = 100 \mid 2 \text{ buys}) = \frac{0.931(0.9)}{0.931(0.9) + 0.069(0.1)} \approx 0.992$$

$$\mathbb{E}[V \mid 2 \text{ buys}] \approx \boxed{\$99.84}$$

▶ *Intuition:* belief convergence is fast — two consecutive buys push you from a 92 prior to within 16 cents of 100. This iterative re-use of yesterday's posterior as today's prior is the Glosten–Milgrom mechanism, the sequential-trade cousin of Kyle's model.

Example 18 — The Mirror Case: a Sell Order

Same prior as Example 11 ($V = 100$ w.p. 0.6, $V = 80$ w.p. 0.4). Informed traders sell w.p. 0.9 if $V = 80$, w.p. 0.1 if $V = 100$. You observe a *sell*. Update your belief.

Solution.

$$\mathbb{P}(V = 80 \mid \text{sell}) = \frac{0.4(0.9)}{0.4(0.9) + 0.6(0.1)} = \frac{0.36}{0.42} \approx 0.857$$

$$\mathbb{E}[V \mid \text{sell}] \approx \boxed{\$82.86}$$

⋆ *Intuition:* exactly symmetric to Example 11 — a sell order is evidence V is low, pulling your belief *down* from 92 toward 80, just as a buy order pulled it up toward 100. Buys and sells are informative in opposite directions by construction.

Example 19 — Full Capstone: Belief, Variance, and Spread Together

Using Example 17's post-2-buy belief ($\mathbb{P}(V{=}100) \approx 0.992$, $\mathbb{E}[V] \approx 99.84$), find the posterior variance and quote $\pm 0.5\sigma$ around the updated mean.

Solution. Two-point variance formula: $\text{Var}(V) = \mathbb{P}(100)\mathbb{P}(80)(100-80)^2$:

$$\text{Var}(V) = 0.992(0.008)(400) \approx 3.24 \Rightarrow \sigma \approx 1.80$$

Half-spread $= 0.5(1.80) = 0.90$:

$$\text{bid} \approx \boxed{\$98.94}, \quad \text{ask} \approx \boxed{\$100.74}$$

▶ *Intuition:* this chains Section C's Bayesian updating straight into Chapter 3.2's σ-based spread convention — notice σ *shrank* as beliefs sharpened (more buys ⇒ more certainty ⇒ tighter justified spread), the opposite of the adverse-selection-driven *widening* you'd apply per-trade in real time.

Example 20 — Mega-Capstone: Winner's Curse Meets Asymmetric Information

You're bidding in a common-value auction, $n = 4$ bidders, your raw estimate = \$205, noise $\sim$ Unif$(-8, 8)$. Separately, you know 25% of rivals have systematically better (lower-noise) private signals, so you apply an extra \$1 defensive widening. Find your final bid.

Solution. Winner's-curse shading (Example 6/7's logic, now $n - 1 = 3$ *other* bidders setting the relevant max):

$$\mathbb{E}[\max \text{ of } 3, \ \text{Unif}(-8, 8)] = -8 + 16 \cdot \frac{3}{4} = \boxed{\$4.00}$$

Total adjustment $= 4.00 + 1.00 = 5.00$:

$$\text{final bid} = 205 - 5 = \boxed{\$200}$$

⋆ *Full-circle intuition:* Section B's winner's curse (you might be wrong about V) and Section C's asymmetric information (others might know more than you) are *separate* risks that stack additively here — a complete competition bid is never just your raw estimate, it's your estimate minus every distinct adverse-selection channel you can identify.

One-tool-tree reminder: a trade is never a random sample of V — it's conditioned on someone wanting it to happen (Section A). Winning an auction or getting a passive fill is exactly this conditioning in disguise (Section B). Bayes' rule turns the whole idea into a precise, updatable number, and order size/direction are themselves signals (Section C). Every capstone in Section D is just these conditioning arguments, applied together and in sequence.